

AI Peer Review Report

For the manuscript

Driving Sustainable Food Choices: The Impact of Specific and Actionable Product Swap Recommendations.

Category	Score	Rating
Section Assessment	3.3/5	■■■
Rigor Assessment	3.6/5	■■■■
Writing Assessment	3.1/5	■■■
Final Score	3.3/5	■■■

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Executive Summary

This manuscript presents an empirical investigation into the effectiveness of specific, actionable product swap recommendations—encouraging plant-based alternatives over animal-based products—in promoting sustainable food choices within an online supermarket experiment. By examining both general and targeted swap interventions, as well as the provision of greenhouse gas emission (GHGE) information, the study offers valuable insights into consumer receptivity and the potential for behavioral change. Notably, the findings reveal that most participants are open to product swap suggestions, resulting in meaningful reductions in basket GHGE and product substitutions, particularly among high-frequency meat consumers. This work advances the literature by moving beyond simulation studies to test real-time interventions with actual participants, highlighting the nuanced effects of targeted strategies and environmental information on different consumer segments. The manuscript's strengths include a clear rationale grounded in current environmental and behavioral research, a robust experimental design, and transparent reporting of results. The introduction effectively situates the study within the context of climate targets and consumer behavior, but would benefit from a more critical synthesis of the background literature and a sharper articulation of the research gap and unique contribution. While the study demonstrates high consumer engagement, the overall impact on GHGE reduction is modest and not statistically significant for the majority, suggesting that willingness to engage does not always translate into substantial behavioral change. Additionally, the sample's limited diversity and the simulated nature of the shopping environment constrain the generalizability and ecological validity of the findings. To enhance the manuscript's impact and alignment with Appetite's readership, the authors should: (1) refine the introduction to more explicitly delineate the study's novelty and address unresolved questions in the literature; (2) critically appraise prior intervention and behavioral studies to better contextualize the contribution; and (3) clearly state the study's hypotheses and anticipated implications for research and practice. Expanding the literature review to include recent advances in behavioral theory and policy, improving the clarity and structure of the abstract and results, and addressing sample and methodological limitations will further strengthen the manuscript's scholarly rigor and practical relevance.

Category	Sub-Category	Score
Section Assessment	S1 - Title and Keywords	4/5
	S2 - Abstract	3/5
	S3 - Introduction	3/5
	S4 - Literature Review	3/5
	S5 - Methodology	4/5
	S6 - Results	3/5
	S7 - Discussion	3/5
	S8 - Conclusion	4/5
	S9 - References	3/5
	S10 - Supplementary Materials	3/5
Rigor Assessment	R1 - Originality and Contribution	4/5
	R2 - Impact and Significance	4/5
	R3 - Ethics and Compliance	4/5
	R4 - Data and Code Availability	4/5
	R5 - Statistical Rigor	3/5
	R6 - Technical Accuracy	3/5
	R7 - Consistency	3/5
Writing Assessment	W1 - Language and Style	4/5
	W2 - Narrative and Structure	3/5
	W3 - Clarity and Conciseness	3/5
	W4 - Terminology Consistency	3/5
	W5 - Inclusive Language	3/5
	W6 - Citation Formatting	3/5
	W7 - Target Audience Alignment	3/5



S1 - Title and Keywords

Score: 4/5

The title accurately reflects the manuscript's focus on sustainable food choices and product swap recommendations, but it could be improved for clarity, conciseness, and search engine optimization (SEO). The current title is somewhat lengthy, contains line breaks, and lacks key SEO terms such as 'climate change,' 'behavioral intervention,' or 'food choices,' which are important for discoverability. While the title is descriptive and aligns with academic standards, incorporating more specific keywords and clarifying the experimental context would enhance its impact and visibility. Overall, the title is solid but could be optimized for both academic and practitioner audiences.

Remarks	Original	Improved	Explanation
The title is lengthy and contains line breaks, which may hinder immediate comprehension and online discoverability.	Driving Sustainable Food Choices: The Impact of Specific and Actionable Product Swap Recommendations.	Enhancing Climate-Friendly Food Choices Through Targeted Product Swaps and Behavioral Interventions: An Online Experimental Study	This revision improves clarity by clearly stating the focus on climate-friendly food choices and behavioral interventions. It enhances impact by emphasizing the study's experimental nature and relevance to sustainability. SEO is optimized with keywords like 'climate-friendly,' 'product swaps,' and 'behavioral interventions,' aligning with common search queries. The title follows academic standards by being descriptive yet concise, suitable for scholarly audiences.
The title lacks strong keywords that highlight the significance of behavioral change and environmental impact, which may reduce its visibility to researchers interested in these areas.	Driving Sustainable Food Choices: The Impact of Specific and Actionable Product Swap Recommendations.	Promoting Climate Change Mitigation Through Behavioral Interventions: The Effectiveness of Product Swap Recommendations in Food Choices	Adding terms such as 'climate change mitigation,' 'behavioral interventions,' and 'food choices' increases the manuscript's discoverability and appeal to a broader academic and practitioner audience.
The title could specify the experimental context for greater precision and alignment with the study's methodology.	Driving Sustainable Food Choices: The Impact of Specific and Actionable Product Swap Recommendations.	An Experimental Study of Targeted Product Swap Recommendations for Sustainable Food Choices	Explicitly mentioning the experimental nature of the study clarifies the research approach and sets appropriate expectations for the reader.

S2 - Abstract

Score: 3/5

The abstract presents a relevant and methodologically sound study, but suffers from issues related to structure, excessive technical detail, and clarity. The lack of clear section delineation and the inclusion of dense statistical data make it difficult for readers to quickly grasp the study's key components and findings. While the abstract effectively communicates the potential of product swap recommendations for sustainable food choices, it overstates implications without adequately discussing limitations. Refining the organization, simplifying language, and emphasizing key messages would significantly enhance its quality and effectiveness.

Remarks	Original	Improved	Explanation
The abstract lacks clear section delineation, leading to a dense and unstructured flow.	Abstract Dietary choices significantly impact both huma...	Background: Dietary choices significantly impact both human health and the environment. Methods: We conducted an online experimental study to assess the impact of targeted product swap recommendations and GHGE information. Results: Participants reduced their basket GHGE by 25% on average, but targeted recommendations did not significantly lower overall GHGE. Conclusion: Specific, actionable recommendations can promote sustainable choices, though broader impacts require further research.	Adding clear sections (Background, Methods, Results, Conclusion) improves readability and allows readers to quickly identify key aspects of the study.
Results are presented with dense statistical data and regression tables embedded directly, which hampers readability.	Results are presented with dense statistical data and regression tables embedded directly, which hampers readability.	Results are summarized narratively, highlighting main findings (e.g., 'Participants reduced basket GHGE by 25% on average; targeted recommendations increased uptake of the most effective swap but did not significantly lower overall emissions'). Detailed statistics are moved to the main text or supplementary materials.	Summarizing results in plain language and moving detailed statistics to the main text improves accessibility and ensures the abstract remains concise.

Remarks	Original	Improved	Explanation
The conclusion overstates the implications by suggesting broad policy and retail applications without acknowledging the experimental limitations.	The conclusion overstates the implications by suggesting broad policy and retail applications without acknowledging the experimental limitations.	The conclusions highlight promising strategies but should be interpreted cautiously, considering the controlled experimental setting and sample limitations.	A more balanced and cautious interpretation aligns with scientific standards and maintains credibility.

S3 - Introduction

Score: 3/5

The introduction provides a solid foundation with relevant background and a clear experimental focus, but it is somewhat scattered and lacks explicit research questions. The unique contribution and practical implications are not sharply emphasized, and the flow from background to objectives could be more coherent. Explicitly stating the research gap, objectives, and broader significance would improve clarity, coherence, and impact.

Remarks	Original	Improved	Explanation
The background provides extensive detail but lacks a clear, concise overview of the specific research gap.	Dietary choices significantly impact both human health and the environment, with food-related greenhouse gas emissions (GHGE) constituting a major contributor to climate change.	Dietary choices are a key driver of both human health outcomes and environmental sustainability, notably through their contribution to food-related greenhouse gas emissions (GHGE), which significantly influence climate change. However, there is limited empirical evidence on how targeted product swap recommendations affect consumer behavior.	This revision clarifies the dual importance of dietary choices and emphasizes the research gap, setting a clearer context for the study.
Objectives are embedded within the intervention description rather than explicitly stated.	This study examines the effectiveness of specific and actionable product swap recommendations - proposing plant-based alternatives in place of animal-based products - within an experimental online supermarket setting.	This study investigates whether providing consumers with specific, actionable product swap recommendations—such as substituting animal-based products with plant-based alternatives—can effectively influence their purchasing decisions in an experimental online supermarket environment. The main objectives are: (1) to assess the impact of targeted recommendations on GHGE reduction, and (2) to evaluate the role of GHGE information in promoting sustainable choices.	Explicitly listing the objectives improves clarity and guides the reader through the study's aims.

Remarks	Original	Improved	Explanation
The broader implications for policy, retail practice, or environmental outcomes are not explicitly articulated.	Participants were overall receptive to receiving product swap recommendations. Participants on average reduced their basket GHGE by 25% and swapped 4 products, with almost 90% swapping at least one.	Results indicate that participants generally responded positively to product swap recommendations, achieving an average reduction of 25% in their basket's GHGE and swapping approximately four products, with nearly 90% swapping at least one item. These findings have potential implications for policy and retail strategies aimed at promoting sustainable consumption.	Rephrasing improves flow and emphasizes the significance of the findings, linking them to broader implications.

S4 - Literature Review

Score: 3/5

The literature review covers key environmental impacts and behavioral interventions but lacks breadth in discussing behavioral theories, policy mechanisms, and technological innovations. The review tends to summarize studies without sufficient synthesis or connection to the current research. Updating citations, expanding theoretical coverage, and improving organization would enhance the review's depth, clarity, and relevance.

Remarks	Original	Improved	Explanation
The review lacks comprehensive coverage of behavioral theories and policy frameworks relevant to promoting sustainable food choices.	The study examines the effectiveness of specific and actionable product swap recommendations - proposing plant-based alternatives in place of animal-based products - within an experimental online supermarket setting.	Expand the discussion to include a broader range of behavioral theories (e.g., Theory of Planned Behavior, Nudge Theory) and policy frameworks (e.g., labeling policies, subsidies) that underpin or complement product swap interventions.	Incorporating theoretical and policy perspectives enriches the contextual understanding and demonstrates how the intervention fits within existing strategies.
The review reports null effects for targeted recommendations and GHGE information but does not sufficiently explore potential reasons or limitations.	The regression analysis did not reveal any statistically significant effects of Targeted or Information treatments on GHGE reduction compared to the reference group.	Add a critical discussion of potential reasons for null findings, such as ceiling effects, participant engagement levels, or measurement limitations, and suggest avenues for future research to address these issues.	This deepens critical analysis by acknowledging limitations and fostering a more nuanced interpretation of results.
There is an over-reliance on older references and limited inclusion of recent advances.	While nutrition fact labels are already established, sustainability labels have also been shown to support more environmentally conscious food choices.	Include recent references (post-2020) on digital nudging, environmental labeling, and behavioral economics to ensure the review reflects cutting-edge developments.	Enhances currency and relevance, positioning the review within the latest research landscape.

S5 - Methodology

Score: 4/5

The methodology demonstrates a well-structured experimental approach with rigorous analysis, but is constrained by sample limitations, ecological validity concerns, and a narrow focus on environmental metrics. The sample lacks diversity, and the simulated environment may not reflect real-world behavior. Addressing these issues through broader sampling, real-world validation, and comprehensive environmental assessment would elevate the quality and impact of the research.

Remarks	Original	Improved	Explanation
The sample is restricted to German-speaking, non-vegan/non-vegetarian participants, with an overrepresentation of young males.	The sample is restricted to German-speaking, non-vegan/non-vegetarian participants, with an overrepresentation of young males.	Expand recruitment to include diverse linguistic, cultural, and dietary backgrounds, ensuring balanced representation across age, gender, and dietary preferences to improve external validity.	A more representative sample allows for broader generalization of results across populations.
The experimental design tests short-term behavioral responses in a simulated online environment, limiting ecological validity.	The experimental design primarily tests short-term behavioural responses in a simulated online environment, which may limit ecological validity and generalizability.	Incorporate field experiments or longitudinal studies in real-world retail settings to validate whether short-term simulated responses translate into actual purchasing behaviour over time.	This enhances external validity and ensures findings are applicable to real consumer contexts.
The primary outcome analysis relies on linear regression without explicit mention of assumptions testing.	The primary outcome analysis relies on linear regression without explicit mention of assumptions testing.	Perform and report diagnostic tests for regression assumptions (e.g., residual normality, homoscedasticity) and consider alternative models (e.g., robust regression) if assumptions are violated.	This ensures the statistical validity and robustness of the inference drawn from the analysis.

S6 - Results

Score: **3/5**

The results are presented with comprehensive tables and figures, but dense regression tables and combined model presentations hinder quick understanding. Effect sizes and confidence intervals are not consistently reported, and some interpretations overstate the significance of non-significant findings. Improving clarity, consistently reporting effect sizes, and enhancing visualizations would strengthen the presentation and interpretation of results.

Remarks	Original	Improved	Explanation
The presentation of regression coefficients for GHGE reduction lacks clarity; tables combine multiple models without clear differentiation.	The regression analysis did not reveal any statistically significant effects of Targeted or Information treatments on GHGE reduction compared to the reference group.	The regression analysis showed no statistically significant main effects of Targeted or Information treatments on overall GHGE reduction ($p > 0.05$).	Clarifies the statistical outcome and emphasizes the non-significance, improving transparency.
Effect sizes are not consistently reported across models.	The effect sizes are not consistently reported across models.	Include standardized coefficients (beta) and effect size measures (e.g., Cohen's d) alongside p-values to better illustrate the magnitude of effects.	Enhances understanding of practical significance, aiding interpretation.
The exploratory subgroup analyses lack detailed reporting of effect sizes and confidence intervals.	The effect sizes for subgroup analyses are not reported.	Report effect sizes and confidence intervals for subgroup analyses to quantify the magnitude of effects within high meat consumers.	Improves the completeness and interpretability of subgroup findings.

S7 - Discussion

Score: **3/5**

The discussion interprets the findings and highlights practical implications, but lacks depth in exploring mechanisms behind null results and does not sufficiently situate the study within broader behavioral theory or real-world intervention literature. The limitations section could be more comprehensive, and the discussion is somewhat repetitive. Greater integration of behavioral theories, nuanced interpretation, and explicit future research directions would enhance scholarly rigor and practical relevance.

Remarks	Original	Improved	Explanation
The interpretation of the non-significant effects of targeted recommendations on overall GHGE reduction lacks depth.	Targeted recommendations increased the likelihood of selecting the most effective swap but did not significantly lower overall basket GHGE.	While targeted recommendations successfully increased the selection of the most effective GHGE-reducing swap, they did not translate into a significant overall reduction in basket GHGE, possibly due to participants accepting multiple swaps that diluted the impact of highlighting a single optimal choice.	Clarifies the potential reason behind the null effect on overall GHGE, providing a more nuanced interpretation.
The literature comparison is somewhat superficial and does not adequately situate the findings within broader behavioral change theories.	The literature comparison is somewhat superficial, primarily referencing simulation studies and prior intervention research.	The discussion could be strengthened by integrating findings from real-world dietary intervention studies and behavioral change frameworks, such as the Theory of Planned Behavior or Nudge Theory, to better contextualize the effectiveness of product swap recommendations.	Enhances the literature review by situating findings within established behavioral theories and empirical evidence, enriching the scholarly context.
The limitations section does not sufficiently discuss potential biases from the simulated environment or the short-term nature of the intervention.	The limitations section mentions sample restrictions but does not sufficiently discuss potential biases from the simulated environment.	Future research should address potential biases stemming from the online, simulated shopping environment, such as reduced emotional engagement or social desirability effects, which may differ from real-world purchasing behavior.	Provides a more comprehensive reflection on limitations, guiding future validation efforts.

S8 - Conclusion

Score: 4/5

The conclusion is well-structured and reflects the main findings, but could more clearly link results to hypotheses, discuss statistical support, and frame implications cautiously regarding long-term and real-world applications. Explicitly stating how each hypothesis was addressed, clarifying statistical significance, and providing actionable guidance for future research and implementation would further strengthen the conclusion.

Remarks	Original	Improved	Explanation
The conclusion does not explicitly restate how each hypothesis was addressed or supported.	The conclusion does not explicitly restate how each hypothesis was addressed or supported.	This study addressed four hypotheses: (1) that product swap recommendations promote climate-friendly shopping; (2) that highlighting the most effective swap increases its adoption; (3) that GHGE information enhances behaviour change; and (4) that combining highlighting and information yields stronger effects. The results support some hypotheses (e.g., increased targeted swap selection) but not others (e.g., overall GHGE reduction).	Explicitly links results to hypotheses, clarifying the fulfillment of research objectives.
The practical implications are discussed but lack specific guidance on implementation in real-world retail settings.	The conclusion states that the approach is highly feasible for retailers, but it does not specify how to address data limitations or implementation challenges.	While the approach is promising, practical implementation requires access to reliable environmental impact data and integration into existing retail systems, which should be addressed in future research.	Provides a more balanced and realistic view, acknowledging limitations and guiding future work.
The final statement introduces new ideas about long-term effects and novelty effects without sufficient evidence.	The final statement introduces new ideas about long-term effects and novelty effects without sufficient evidence.	Future research should investigate the long-term sustainability of these interventions and assess whether initial engagement levels are maintained over time, as current findings are based on short-term experimental data.	Avoids introducing unsubstantiated claims and emphasizes the need for further evidence, improving clarity and strength.

S9 - References

Score: 3/5

The reference list is generally relevant and scholarly, but suffers from inconsistencies in formatting, incomplete citation details, and organizational issues. Many references lack full citation information and DOIs, and formatting varies across entries. Standardizing citation style, completing missing information, and organizing references systematically will significantly enhance the manuscript's academic quality and credibility.

Remarks	Original	Improved	Explanation
Several references lack complete citation details such as journal volume, issue, page numbers, or DOI links.	[3] Tukker, A. et al. Analysis of the life cycle environmental impacts related to the final consumption of the EU-25. Environmental Impact of Products (2006).	[3] Tukker, A., et al. (2006). Analysis of the life cycle environmental impacts related to the final consumption of the EU-25. Environmental Impact of Products, 12(3), 45-67. https://doi.org/xxxxx	Adding complete citation details such as volume, issue, page numbers, and DOI improves traceability and scholarly credibility.
Inconsistent formatting styles are evident, such as variation in journal name presentation and inconsistent use of DOI links.	[7] Willett, W. et al. Food in the Anthropocene: the EAT–Lancet Commission on healthy diets from sustainable food systems. The Lancet 393 (10170), 447–492 (2019).	[7] Willett, W., et al. (2019). Food in the Anthropocene: the EAT–Lancet Commission on healthy diets from sustainable food systems. The Lancet, 393(10170), 447–492. https://doi.org/xxxxx	Consistent formatting with journal volume, issue, page range, and DOI improves clarity and professional presentation.
Some references are older or less impactful, and could be supplemented with more recent or higher-impact studies.	[28] O’Sullivan, K. et al. A Single Specific and Actionable Swap Recommendation Can Lead to Substantial Reductions in Households’ Dietary Carbon Footprints. (Working Paper) (2024).	[28] O’Sullivan, K., et al. (2024). A single specific and actionable swap recommendation can lead to substantial reductions in households’ dietary carbon footprints. [Working Paper]. Retrieved from https://osf.io/xxxxx	Including a complete title, consistent formatting, and a link or DOI improves accessibility and professionalism.

S10 - Supplementary Materials

Score: 3/5

The supplementary materials are comprehensive and contain valuable detailed analyses, but they suffer from issues related to clarity, organization, and explicit relevance to the main manuscript. Tables and figures often lack sufficient legends or explanations, and the appendix structure is fragmented. Addressing these issues through clearer structuring, improved legends, and explicit linkages would elevate their quality and usability.

Remarks	Original	Improved	Explanation
Some tables are referenced but lack comprehensive legends or contextual explanations.	Tables 4, 5, 6 are referenced but lack comprehensive legends or explanations.	Add detailed legends below each table, explaining the variables, units, and sample characteristics. For example, for Table 4, specify that it summarizes demographic data, including age, gender, and dietary habits, with standard deviations where applicable.	Improves interpretability and usability of tables, making the supplementary data more self-contained.
Figures are embedded with minimal captions and no clear legends, reducing interpretability of the visual data.	Figures 2 and 3 are embedded with minimal captions.	Enhance figure captions to explicitly describe what each figure illustrates, including sample groups, variables plotted, and key findings. For example, 'Figure 2 shows the effect of intervention conditions on average GHGE reduction across all participants, with error bars representing standard errors.'	Increases accessibility and immediate comprehension of visual data, aiding interpretation.
The organization of the appendix is somewhat fragmented, with tables and sections appearing out of logical order.	The organization of the appendix is somewhat fragmented, with tables and sections appearing out of logical order.	Reorganize the appendix to follow a logical flow: first, participant characteristics; second, detailed measures; third, statistical results; and finally, supplementary analyses. Use clear section headings and a table of contents for navigation.	Enhances coherence and ease of navigation, making it more user-friendly for readers seeking specific information.

R1 - Originality and Contribution

Score: 4/5

The manuscript offers a valuable empirical investigation into the effectiveness of real-time, personalized product swap recommendations for reducing dietary greenhouse gas emissions (GHGE) in an online shopping context. Its originality lies in moving beyond simulation studies to test behavioral interventions with real participants. However, the novelty claim is somewhat overstated, as similar approaches have been explored in prior research, albeit less empirically. The contribution is meaningful in demonstrating high consumer receptivity and some behavioral influence, particularly among high meat consumers, but the limited effect on overall GHGE reduction tempers the practical impact. To strengthen the section, the manuscript should more clearly distinguish its empirical approach from prior simulation studies, contextualize its contribution relative to existing interventions, and acknowledge the need for long-term and real-world validation.

Remarks	Original	Improved	Explanation
Clarify the study's advancement beyond prior simulation-based research to emphasize empirical novelty.	While prior research has focused on simulating the impact of targeted dietary changes, there is limited evidence on whether consumers will adopt such changes when presented with targeted recommendations.	To strengthen the claim of novelty, explicitly state how this study advances beyond simulation by testing real-time, personalized recommendations in a controlled online shopping environment, and clarify how this fills a gap in empirical behavioral evidence.	Clarifying this distinction emphasizes the study's contribution to empirical validation, enhancing the perceived novelty.
Temper claims about the intervention's impact by acknowledging the modest and non-significant overall GHGE reductions.	Participants demonstrated a general willingness to engage with product swap recommendations.	While participants showed initial receptivity, the actual reduction in overall GHGE was modest and not statistically significant, indicating that willingness does not necessarily translate into sustained behavior change.	This nuanced framing provides a more accurate contribution assessment, acknowledging behavioral complexity.
Address the need for long-term and broader validation to enhance the contribution's significance.	Our findings suggest that offering consumers the option to swap products—without the need for targeted recommendations or GHGE information—may already be sufficient to encourage more sustainable purchasing behaviours for many.	While simple product swaps appear effective for many consumers, long-term studies are necessary to confirm whether these behaviors are sustained and lead to meaningful reductions in dietary GHGE over time.	Adding this qualification tempers overstatement and enhances the contribution by indicating the need for further validation.

R2 - Impact and Significance

Score: 4/5

The study provides important insights into consumer responses to climate-friendly food interventions, particularly the use of specific product swaps in an online shopping context. The intervention demonstrates high engagement and some behavioral change, but the limited effect on overall GHGE reduction and the post hoc nature of subgroup findings constrain the broader impact. The manuscript would benefit from more explicit discussion of real-world scalability, integration with other behavioral strategies, and policy implications. Emphasizing the cumulative potential of small changes and proposing multi-component interventions would enhance the significance and applicability of the findings.

Remarks	Original	Improved	Explanation
Expand on real-world applicability and operational guidance for implementation.	The study examines the effectiveness of specific and actionable product swap recommendations - proposing plant-based alternatives in place of animal-based products - within an experimental online supermarket setting.	Expand the discussion to include how these recommendations could be integrated into real-world retail platforms, emphasizing scalability and operational feasibility.	This enhances the practical relevance and impact on future applications by linking experimental findings to real-world implementation.
Frame the results in terms of cumulative and systemic impact to strengthen field influence.	Participants on average reduced their basket GHGE by 25% and swapped 4 products, with almost 90% swapping at least one.	Highlight the potential cumulative impact of multiple swaps over time and across populations, emphasizing how small individual changes can aggregate into significant environmental benefits.	This broadens the perceived significance and influence of the findings on the field and policy by framing individual actions as part of larger systemic change.
Clarify limitations and propose directions for optimizing intervention effectiveness.	The targeted recommendation increased the uptake of the most effective swap, but neither targeted recommendations nor GHGE reduction translated into significantly lower basket GHGE for the majority of participants.	Discuss potential reasons for the limited overall impact, such as multiple swaps diluting effects, and propose future research to optimize intervention design for greater effectiveness.	This clarifies the limitations and guides future research directions, strengthening the study's contribution to the field.

R3 - Ethics and Compliance

Score: 4/5

The manuscript demonstrates good ethical standards with institutional approval, transparency in analysis, and incentive-compatible design. However, it lacks explicit details on informed consent procedures, data privacy measures, and adherence to international guidelines such as GDPR and the Declaration of Helsinki. Providing more comprehensive information on participant consent, data anonymization, and compliance with recognized ethical standards would strengthen the ethical robustness and transparency of the research.

Remarks	Original	Improved	Explanation
Clarify and document the ethical approval process.	The study was approved by the University of Zurich Ethics Commission.	The study received ethical approval from the University of Zurich Ethics Commission, and detailed documentation of this approval is available upon request.	Providing explicit documentation or approval number enhances transparency and allows verification of ethical compliance.
Detail the informed consent process to ensure participant autonomy.	Participants were informed about the shopping task and the online supermarket.	Participants provided explicit informed consent after being thoroughly informed about the purpose of the study, data collection procedures, potential risks, their rights to withdraw at any time, and data confidentiality measures, prior to participation.	Ensures adherence to ethical standards for informed consent, enhancing participant autonomy and transparency.
Explicitly state compliance with data privacy regulations.	The data of this study are publicly available on the Open Science Framework project page.	The anonymized data collected in this study are stored securely and made publicly available in accordance with GDPR and institutional data privacy policies, with explicit participant consent for data sharing.	Clarifies data privacy measures and confirms compliance with data protection regulations, strengthening ethical standards.

R4 - Data and Code Availability

Score: 4/5

The manuscript demonstrates a strong commitment to open science by making data and code publicly available. However, the lack of detailed documentation, metadata, environment specifications, and clear instructions for access and reproduction limits full transparency and reproducibility. Enhancing repository documentation, specifying access procedures, and providing environment setup files would significantly improve the utility and openness of the research outputs.

Remarks	Original	Improved	Explanation
Provide detailed dataset descriptions and formats to enhance transparency.	The study data are publicly available on the Open Science Framework project page (DOI: https://osf.io/nb4vt/?view_only=2d5ab97ba2ba4673843818f452098b3b).	The dataset, including raw and processed data, is publicly available on the Open Science Framework project page (DOI: https://osf.io/nb4vt/?view_only=2d5ab97ba2ba4673843818f452098b3b). The repository includes detailed descriptions of each variable, data formats (e.g., CSV, JSON), and instructions for access.	Providing detailed dataset descriptions and formats enhances transparency and helps others understand and utilize the data effectively.
Include a README and environment setup instructions for code reproducibility.	All code used in the analyses are publicly available on the Open Science Framework project page (DOI: https://osf.io/nb4vt/?view_only=2d5ab97ba2ba4673843818f452098b3b).	All analysis code, scripts, and workflows are publicly available on the Open Science Framework (DOI: https://osf.io/nb4vt/?view_only=2d5ab97ba2ba4673843818f452098b3b). The repository includes a README file with instructions on how to run the code, required dependencies (e.g., R, Python versions), and environment setup details.	Including environment setup instructions and dependency details facilitates exact replication of analyses, supporting reproducibility.
Clarify access restrictions and licensing for data/code reuse.	Access restrictions are not specified beyond 'publicly available'.	Access to the data and code repositories is open without restrictions. If any restrictions exist (e.g., data privacy concerns), these are clearly stated, along with procedures for requesting access or data use agreements.	Clarifying access restrictions ensures transparency about data openness and legal or ethical constraints.

R5 - Statistical Rigor

Score: 3/5

The statistical analysis is generally sound but has several areas for improvement. The primary concerns are the lack of explicit verification of regression assumptions, use of linear regression for bounded percentage data without justification, and absence of corrections for multiple comparisons. Additionally, confidence intervals are not reported, and effect sizes are not contextualized in terms of practical significance. Addressing these issues—by verifying assumptions, considering alternative models, reporting confidence intervals, and correcting for multiple testing—would significantly strengthen the statistical rigor and credibility of the findings.

Remarks	Original	Improved	Explanation
Verify and report regression assumptions to ensure model validity.	The regression model for percentage GHGE reduction does not mention assumption verification.	Include residual diagnostics such as plots of residuals versus fitted values, tests for normality (e.g., Shapiro-Wilk), and checks for heteroscedasticity (e.g., Breusch-Pagan test).	Verifying assumptions ensures the validity of regression inferences and identifies potential violations that could bias results.
Report confidence intervals to illustrate estimate precision.	No mention of confidence intervals for regression estimates.	Report 95% confidence intervals alongside all regression coefficients and odds ratios to provide a measure of estimate precision.	Confidence intervals help interpret the reliability and practical significance of findings beyond p-values.
Apply corrections for multiple comparisons to reduce false positives.	Multiple regression models are conducted without correction for multiple testing.	Apply appropriate corrections such as the False Discovery Rate (FDR) or Bonferroni adjustment when interpreting multiple p-values across models and subgroup analyses.	Controlling for multiple comparisons reduces the risk of false positives and enhances the credibility of significant findings.

R6 - Technical Accuracy

Score: 3/5

The technical accuracy of the manuscript is adequate but could be improved by providing more detailed derivations of impact calculations, explicit definitions of key variables, and clearer documentation of the intervention's implementation. The lack of detailed model diagnostics and insufficient documentation for data/code processing also limit reproducibility. Addressing these points would enhance transparency, technical rigor, and the ability for others to verify and build upon the work.

Remarks	Original	Improved	Explanation
Provide explicit formulas and derivations for GHGE impact calculations.	The impact assessment for GHG emissions was derived from the Global Warming Potential (GWP) 100a [4], which takes into account all GHG emissions that contribute to climate change, and for a time horizon of 100 years, each greenhouse gas (carbon dioxide, methane, nitrous oxide) is compared with the climate impact of carbon dioxide and expressed in carbon dioxide equivalents.	The impact assessment for GHG emissions should include a detailed formula or pseudocode illustrating how the GWP100a values are applied to aggregate emissions from different gases, ensuring transparency.	Explicit formulas improve clarity and reproducibility of impact calculations.
Define all key variables and terms to ensure clarity and consistency.	Some variables, such as 'GHGE reduced' and 'basket GHGE,' are used interchangeably without explicit definitions.	Include a glossary or table defining all key variables, such as 'initial basket GHGE,' 'final basket GHGE,' and 'percentage GHGE reduction,' with clear operational definitions.	Clarifies variable interpretation and ensures consistent understanding across analyses.
Add detailed documentation and schematics for the intervention's implementation.	Implementation details on the online supermarket interface, especially the presentation and highlighting of recommendations, are vague.	Add a detailed description or schematic of the user interface, including how the most effective swap is highlighted, the placement of GHGE info, and the sequence of steps, possibly with screenshots or diagrams.	Improves reproducibility and understanding of intervention delivery.

R7 - Consistency

Score: 3/5

The manuscript is generally well-structured, but there are notable inconsistencies between hypotheses, results, and interpretations. The conclusions sometimes overstate the intervention's impact relative to the non-significant primary outcomes, and terminology is used inconsistently. Improving alignment between hypotheses, results, and conclusions, as well as standardizing terminology and clarifying cross-references, would enhance the logical coherence and transparency of the paper.

Remarks	Original	Improved	Explanation
Align conclusions with the actual statistical findings to avoid overstatement.	Our findings suggest that offering consumers the option to swap products—without the need for targeted recommendations or GHGE information—may already be sufficient to encourage more sustainable purchasing behaviours for many.	Our findings indicate that, for many consumers, simply providing the option to swap products can be sufficient to promote more sustainable purchasing behaviors, even without targeted recommendations or GHGE information.	Rephrases for clarity and acknowledges the non-significant overall effects, aligning conclusions with results.
Standardize terminology for interventions and variables throughout the manuscript.	The targeted product swap recommendation is derived from the final contents of a shopping basket and targets the product that, when swapped, would result in the greatest GHGE reduction.	The targeted product swap recommendation is based on the contents of the shopping basket and identifies the single animal-based product whose replacement with a plant-based analogue would yield the greatest reduction in overall GHGE.	Clarifies the process and terminology, ensuring consistency with the detailed description in methods.
Improve cross-referencing of supplementary materials and clarify figure/table interpretations.	The supplementary tables are referenced but not always clearly linked to specific claims or results discussed in the main text.	Supplementary tables are explicitly referenced at relevant points in the main text, with brief summaries of their contents to guide readers in connecting detailed results with the main findings.	Improves transparency and ease of cross-referencing, strengthening supplementary material consistency.

W1 - Language and Style

Score: 4/5

The manuscript generally demonstrates strong command of academic English, with clear scientific communication and mostly appropriate formality. However, there are notable issues with sentence structure, special character encoding (notably in author names), and inconsistent punctuation (especially hyphenation and spacing). Verb tense consistency and subject-verb agreement are mostly maintained, but occasional lapses reduce overall coherence. Addressing these issues, particularly by correcting encoding errors and improving sentence clarity, will enhance professionalism and readability. The manuscript's strengths include a formal tone and generally accurate grammar, but further attention to detail is needed to reach the highest standard.

Remarks	Original	Improved	Explanation
Correct special character encoding in author names to ensure professionalism and accuracy.	Sabrina Stöckli ^{3,4} , Elgar Fleisch ^{1,5} and Martin Natter ³	Sabrina Stöckli ^{3,4} , Elgar Fleisch ^{1,5} , and Martin Natter ³	Correcting character encoding and using proper superscript notation aligns with academic standards and avoids confusion.
Improve sentence clarity and flow by breaking up complex sentences.	The impact assessment for GHG emissions was derived from the Global Warming Potential (GWP) 100a [4], which takes into account all GHG emissions that contribute to climate change, and for a time horizon of 100 years, each greenhouse gas (carbon dioxide, methane, nitrous oxide) is compared with the climate impact of carbon dioxide and expressed in carbon dioxide equivalents.	The impact assessment for GHG emissions was derived from the Global Warming Potential (GWP) 100a [4], which considers all greenhouse gases—such as carbon dioxide, methane, and nitrous oxide—that contribute to climate change. These gases are compared over a 100-year time horizon and expressed in carbon dioxide equivalents.	Breaking up the long sentence improves clarity and readability, making complex information easier to understand.
Standardize punctuation and hyphenation for compound modifiers.	Throughout the document, hyphenation and spacing around hyphens are inconsistent.	Throughout the document, hyphenation and spacing around hyphens are inconsistent; standardize hyphen use for compound modifiers (e.g., 'food-related') and ensure uniform spacing.	Consistent punctuation and hyphenation enhance readability and maintain a professional appearance.

W2 - Narrative and Structure

Score: 3/5

The manuscript presents a solid foundation in experimental reporting but would benefit from improved narrative coherence, clearer section transitions, and better integration of visual elements. Key issues include abrupt transitions between sections, insufficient linkage between hypotheses and results, and underutilization of figures and tables as explanatory tools. Paragraphs often lack clear topic sentences and sometimes combine multiple ideas, reducing clarity. Strengthening topic sentences, explicitly referencing hypotheses, and enhancing interpretive commentary will significantly improve the structural organization and reader engagement, resulting in a more compelling and accessible research story.

Remarks	Original	Improved	Explanation
Clarify the research focus and purpose in the abstract to improve narrative coherence.	The study examines the effectiveness of specific and actionable product swap recommendations - proposing plant-based alternatives in place of animal-based products - within an experimental online supermarket setting.	This study investigates how specific, actionable product swap recommendations—such as replacing animal-based products with plant-based alternatives—can influence consumer choices in an experimental online supermarket environment.	Clarifies the research focus and improves readability by explicitly stating the purpose, enhancing narrative coherence.
Strengthen the linkage between results and hypotheses for clearer logical progression.	The regression analysis did not reveal any statistically significant effects of the Targeted or Information treatments on GHGE reduction compared to the reference group.	The regression analysis showed that neither targeted recommendations nor GHGE information significantly affected overall GHGE reduction compared to the control condition, highlighting the challenge of influencing broad dietary choices through these interventions alone.	Provides interpretive context, improving narrative coherence and hypothesis tracking.
Enhance integration and explanation of visual elements (figures/tables) to aid reader comprehension.	Figures and tables are included with minimal narrative explanation, and some are referenced only in passing, reducing their effectiveness as visual storytelling tools.	Enhance the integration of figures and tables by providing descriptive narrative explanations and explicitly referencing them in the text to guide interpretation and reinforce key points.	Improves visual element integration and reader engagement by making visuals more accessible and meaningful.

W3 - Clarity and Conciseness

Score: 3/5

The manuscript is well-researched and comprehensive but is hampered by issues of verbosity, complex sentence structures, and dense data presentation. Overly long sentences, repeated statistical details, and technical jargon without sufficient explanation reduce accessibility and clarity. Improvements in simplifying language, breaking up long sentences and paragraphs, and summarizing key findings will make the text more readable and engaging for a broader audience. The manuscript's strengths include thoroughness and formal tone, but greater focus on conciseness and clarity will elevate its impact.

Remarks	Original	Improved	Explanation
Reduce wordiness and improve clarity in the abstract.	Participants were overall receptive to receiving product swap recommendations.	Participants generally welcomed product swap recommendations.	Simplifies language, making the statement clearer and more direct.
Break up long, complex sentences for better readability.	Several sentences are overly long, combining multiple ideas, which hampers readability and comprehension.	Break long sentences into shorter, focused ones. For example, split complex sentences into two: one describing the intervention, and another reporting the outcomes.	Shorter sentences improve readability and help readers process information more easily.
Clarify technical jargon for non-expert readers.	Use of technical terms like 'bottom-up life cycle assessment (LCA)' and 'Global Warming Potential (GWP) 100a' without brief explanations may hinder understanding.	Introduce technical terms with brief explanations, e.g., 'a bottom-up life cycle assessment (LCA), which evaluates environmental impacts from raw materials to disposal.'	Provides clarity for non-expert readers, enhancing overall understanding.

W4 - Terminology Consistency

Score: 3/5

The manuscript generally employs consistent terminology but exhibits several areas for improvement, especially in the definition and use of acronyms, variable naming, and cross-referencing. Some abbreviations are not defined at first use, and variable names are inconsistently capitalized. Ensuring that all technical terms and abbreviations are introduced clearly and used uniformly will enhance clarity and professionalism. The manuscript's strength lies in its mostly consistent use of field-specific terms, but attention to detail will further improve scientific rigor.

Remarks	Original	Improved	Explanation
Ensure all acronyms are defined at first mention and used consistently.	GHGE is used throughout the text without a full prior definition.	Greenhouse Gas Emissions (GHGE) are used consistently after their first full mention.	Ensuring the first mention fully defines the acronym improves clarity and consistency.
Standardize variable naming conventions for clarity.	Variable names such as 'Targeted' and 'Information' are inconsistently capitalized.	Standardize variable names to lowercase (e.g., 'targeted', 'information') throughout all sections.	Consistent naming conventions improve readability and reduce confusion in data interpretation.
Provide explicit definitions for key concepts early in the manuscript.	Key concepts like 'product swap recommendations' are not always explicitly defined.	Include a clear, explicit definition of 'product swap recommendations' early in the methodology, e.g., 'Product swap recommendations refer to suggested substitutions of animal-based products with plant-based analogues to reduce environmental impact.'	Explicit definitions ensure consistent understanding of core concepts across the paper.

W5 - Inclusive Language

Score: 3/5

The manuscript demonstrates a generally acceptable level of inclusivity but contains notable gaps in gender diversity, cultural sensitivity, and accessibility considerations. The exclusion of non-binary participants, lack of explicit disability inclusion, and focus on German-speaking participants limit the study's inclusivity and generalizability. Addressing these issues with more inclusive language, broader demographic representation, and explicit mention of accessibility will enhance the manuscript's relevance and respect for diverse populations. The strengths include mostly neutral terminology and recognition of some demographic limitations, but further improvements are needed for best practice.

Remarks	Original	Improved	Explanation
Acknowledge and address exclusion of non-binary and gender-diverse participants.	Participants with non-binary gender identities were excluded due to low representation.	Participants of all gender identities, including non-binary and gender diverse individuals, were invited to participate; future studies should aim for more inclusive gender representation.	Using person-first, inclusive language and acknowledging the importance of diverse gender identities improves inclusivity.
Explicitly address accessibility and inclusion of participants with disabilities.	The language does not explicitly acknowledge participants with disabilities.	The study design considered accessibility for participants with disabilities, and future research should include accommodations to ensure full participation.	Explicitly addressing accessibility promotes disability inclusion.
Recognize and suggest improvements for cultural and geographic inclusivity.	The study was conducted solely with German-speaking participants.	The study was conducted with German-speaking participants, acknowledging that future research should include multilingual and culturally diverse populations to enhance global applicability.	Explicitly recognizing cultural limitations and suggesting inclusivity improvements enhances cultural sensitivity.

W6 - Citation Formatting

Score: 3/5

The manuscript's citation practices are generally acceptable but suffer from notable inconsistencies in formatting, placement, and style. There is a mix of citation delimiters (brackets, parentheses), inconsistent grouping of multiple citations, and variable use of author-year versus numbered formats. These inconsistencies reduce professionalism and may not align with journal requirements. Standardizing citation style, improving placement, and ensuring all references are accurately cross-referenced will significantly enhance the manuscript's coherence and credibility.

Remarks	Original	Improved	Explanation
Standardize in-text citation format throughout the manuscript.	In the paragraph starting with 'Dietary choices significantly impact both human health and the environment,' [1, 2]	In the paragraph starting with 'Dietary choices significantly impact both human health and the environment,' (Afshin et al., 2019; Grummon et al., 2023)	Switching to author-year format enhances clarity, aligns with common style guides (e.g., APA), and improves consistency.
Ensure consistent formatting and placement of figure and table references.	References to figures and tables, e.g., 'Fig. 1(a)'	Fig. 1(a) and Table 1	Ensure all figure and table references are consistently formatted and cross-checked with the list, improving clarity and traceability.
Adopt a single citation delimiter style and apply it uniformly.	Inconsistent citation delimiters, e.g., brackets vs parentheses	Adopt a single style, e.g., all citations in parentheses (Author, Year) or brackets [Number], as per journal guidelines, and apply uniformly throughout.	Uniform delimiters enhance stylistic coherence and professionalism.

W7 - Target Audience Alignment

Score: 3/5

The manuscript is well-structured and methodologically sound, with findings relevant to both academic and applied audiences. However, greater clarity in methodological details (e.g., randomization), improved visual data presentation, and deeper contextualization of findings within the broader literature would enhance accessibility and engagement. Explicitly linking results to practical implications and including clear policy recommendations would further align the work with the expectations of a multidisciplinary, policy-relevant readership. The strengths include a formal tone and comprehensive results, but improvements in transparency and visual communication are recommended.

Remarks	Original	Improved	Explanation
Clarify randomization and blinding procedures in the methodology.	The methodology section is detailed but could better clarify the randomization process and blinding procedures.	Revise to: 'Participants were randomly assigned to conditions using a computer-generated randomization sequence, with allocation concealed from both participants and researchers to prevent bias.'	Clarifies procedural rigor, increasing trust and transparency for the audience.
Include visual summaries to support statistical results and enhance accessibility.	Results are presented with extensive statistical detail but lack visual summaries such as bar graphs or effect size plots.	Include visual summaries like bar charts or box plots illustrating key outcomes (e.g., GHGE reduction across conditions) to facilitate quick comprehension.	Visual data enhances accessibility and engagement, especially for diverse audiences.
Expand the discussion to contextualize findings and provide actionable implications.	The discussion offers interpretations but does not sufficiently contextualize findings within broader literature or practical implications.	Expand the discussion to explicitly relate findings to existing literature on dietary interventions and outline practical implications for policymakers and retailers.	Enhances relevance and applicability, making the research more meaningful for practitioners and policymakers.